Using Generative AI to Summarize and Compare SEC 10-K Filings for Investment Research

Investment research is a time-consuming and labor-intensive process. Analysts often need to review annual reports that can exceed 100 pages, and in some industries, such as banking, 10-K filings may be 300 pages or longer. Carefully reading and analyzing such a large volume of information requires significant effort and sustained attention. By the time an analyst completes this process, which may take several days, market prices may have already adjusted, and potential investment opportunities may no longer appear undervalued. Additionally, effective investment analysis requires strong knowledge of accounting rules and financial statement interpretation.

Generative artificial intelligence (GAI) offers an opportunity to improve this process. Large language models can rapidly process lengthy textual documents and extract relevant information within seconds. Unlike human analysts, these models do not experience fatigue or loss of concentration. As a result, generative AI has the potential to make investment research more efficient by reducing the time required to analyze complex regulatory filings and allowing analysts to focus on higher-level decision making.

Updated Problem Statement

The original version of this project focused on identifying companies with durable competitive advantages at an early stage using publicly available data. However, after further review and instructor feedback, it became clear that this approach was not viable. Many well-known companies, such as Amazon, Google, Microsoft, and Nvidia, became public only after their early development stages, meaning relevant historical data is limited or unavailable. In addition, that approach relied more heavily on predictive analytics rather than generative AI.

To better align with the goals of this course, the project has been revised to focus on the application of generative AI for investment research tasks. Specifically, the project examines how large language models can be used to summarize, extract, and compare information from SEC 10-K filings to support more efficient and informed investment analysis.

Generative AI Approach

This project will use a large language model (LLM) as the core generative AI technology. LLMs are well suited for processing natural language documents, such as regulatory filings, earnings transcripts, and financial disclosures. These models can summarize large volumes of text, extract key themes, and compare information across multiple documents.

The use of an LLM is appropriate because investment research often involves synthesizing qualitative information from diverse sources. Tasks such as identifying business risks, understanding management strategy, and comparing companies across industries rely heavily on textual analysis rather than numerical prediction.

Prompt Experiments

To evaluate whether generative AI can effectively support investment research, the following prompt experiments will be conducted:

Prompt 1 – Executive Summary

Summarize this 10-K filing in plain English for a long-term investor. Focus on the business model, revenue sources, and competitive position.

Prompt 2 – Risk Extraction

Extract and summarize the top five business risks mentioned in this 10-K. Explain why each risk matters to investors.

Prompt 3 – Strategic Focus

What strategic priorities does management emphasize in this 10-K? Support your answer with examples from the text.

Prompt 4 – Year-over-Year Comparison

Compare this year's 10-K filing to the previous year's filing. Identify any major changes in strategy, risks, or financial focus.

Prompt 5 – Company Comparison

Compare the 10-K filings of Company A and Company B. Highlight key similarities and differences in business model, risks, and competitive positioning.

Initial experiments suggest that these prompts can generate concise and relevant summaries, although some responses may lack financial nuance or require additional validation by a human analyst. Overall, the prompts demonstrate potential for improving research efficiency.

Fine-Tuning Strategy

If this project were expanded beyond the initial scope, fine-tuning could be performed using a curated dataset of historical 10-K filings. Training examples could focus on sections related to business risks, management discussion, and competitive positioning. Fine-tuning would aim to reduce generic responses and improve the model's ability to generate company-specific insights relevant to investment analysis.

Limitations and Future Work

Despite its potential benefits, this approach has limitations. Large language models may produce hallucinations or generate inaccurate interpretations of financial information. Context length limitations may also restrict the amount of text that can be analyzed at once. In addition, models rely solely on publicly available information and cannot account for undisclosed risks or future regulatory changes.

Future work may include improving prompt design, experimenting with document chunking strategies, and incorporating human review into the analysis process. These enhancements could help make generative AI a more reliable tool for investment research.